Python Turtle: Draw with Code

Make shapes, patterns and art with just a few lines.

Python | Beginner | Ages 8–15

The Big Idea

- Turtle graphics lets you draw by giving a little turtle commands.
- The turtle starts in the middle of the screen, facing right.
- Wherever the turtle moves with its pen down, it draws a line.
- Loops turn a few commands into amazing patterns.
- It comes with Python, so there is nothing to install.

Basic Commands

- `forward(100)` and `backward(100)` move the turtle.
- `left(90)` and `right(90)` turn it on the spot.
- `penup()` lifts the pen, `pendown()` puts it down.
- `goto(x, y)` jumps to a position. `home()` goes back to the middle.
- `circle(50)` draws a circle with radius 50.

Style Commands

- `color("red")` sets the pen colour. `pensize(5)` makes it thicker.
- `begin_fill()` and `end_fill()` fill a shape with colour.
- `speed(0)` is fastest, `speed(1)` is slowest.
- `bgcolor("black")` sets the background (on the screen object).
- `hideturtle()` hides the arrow when you are finished.

The Angle Secret

- To draw a shape with n sides, turn 360 ÷ n degrees each time.
- Square: 4 sides and 90 degrees. Triangle: 3 sides and 120 degrees. Hexagon: 6 sides and 60 degrees.
- A star with 5 points uses a turn of 144 degrees.
- Add `turtle.done()` at the end so the window stays open.

💻 Code Examples

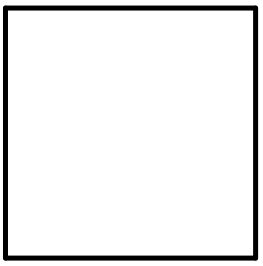
A square python

```python
import turtle
t = turtle.Turtle()

for i in range(4):
    t.forward(100)
    t.left(90)

turtle.done()
```

▶ What you see:



Four sides, four 90 degree turns.

A coloured star python

```python
import turtle
t = turtle.Turtle()
t.color("red", "yellow")

t.begin_fill()
for i in range(5):
    t.forward(120)
    t.right(144)
t.end_fill()

turtle.done()
```

▶ What you see:



begin_fill and end_fill colour the inside of the star.

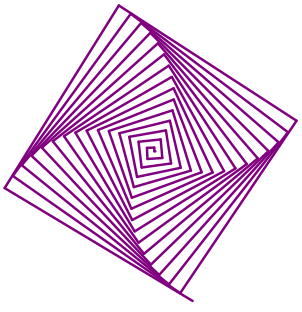
A spiral python

```python
import turtle
t = turtle.Turtle()
t.color("purple")
t.pensize(2)

for i in range(60):
    t.forward(i * 3)
    t.left(91)

turtle.done()
```

▶ What you see:



The line gets longer each time, so it spirals outward.

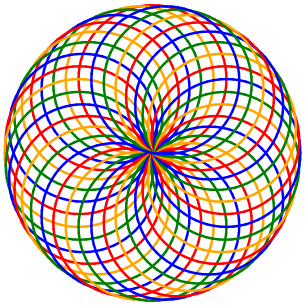
A flower of circles python

```python
import turtle
t = turtle.Turtle()
colours = ["red", "orange", "green", "blue"]

for i in range(36):
    t.color(colours[i % 4])
    t.circle(60)
    t.left(10)

turtle.done()
```

▶ What you see:



36 circles, each turned 10 degrees, make a flower pattern.

Move without drawing python

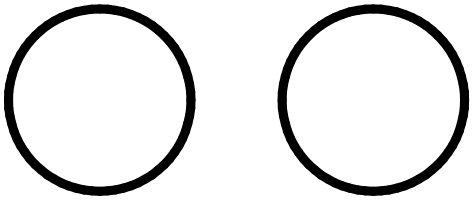
```python
import turtle
t = turtle.Turtle()
t.pensize(4)

t.penup()
t.goto(-80, 0)
t.pendown()
t.circle(40)

t.penup()
t.goto(40, 0)
t.pendown()
t.circle(40)

turtle.done()
```

▶ What you see:



penup and goto let you jump to a new spot without a line.

📚 Key Terms

Turtle
The little arrow that draws on the screen.

Fill
Colouring the inside of a shape.

Pen
The part of the turtle that makes lines.

Coordinates
The x and y position of the turtle.

Heading
The direction the turtle faces.

Screen
The window the turtle draws in.

Radius
The distance from the centre of a circle to its edge.

Loop
Repeating commands to draw patterns.

Angle
How much the turtle turns, in degrees.

Event loop
turtle.done() keeps the window open.

🧮 Quick Reference

```
import turtle
t = turtle.Turtle()
t.forward(100)   t.left(90)
t.penup()   t.goto(0, 50)   t.pendown()
t.color("red")   t.pensize(3)
t.begin_fill()   t.end_fill()
t.circle(50)
turtle.done()
```

✏️ Worked Examples

Q: Draw a triangle.
A: A triangle has 3 sides, so turn 360 / 3 = 120 degrees. for i in range(3): t.forward(100); t.left(120).

Q: Draw a hexagon.
A: 6 sides, so turn 360 / 6 = 60 degrees. for i in range(6): t.forward(80); t.left(60).

Q: Make the pen thicker and green.
A: Use t.color("green") and t.pensize(6) before drawing.

🚫 Common Mistakes

- Forgetting `import turtle`.
- Naming your file `turtle.py`, which stops the module from loading.
- Forgetting `turtle.done()`, so the window closes straight away.
- Mixing up `left` and `right`, or using the wrong angle for a shape.
- Forgetting `pendown()` after moving, so nothing is drawn.

👨‍👩‍👧 Parent Corner

- Ask your child to act as the turtle. Walk a square in the room and say the commands.
- Ask: What happens if we change 90 to 80? Let them predict, then test.
- Connect to maths: angles in a full turn add to 360.
- Encourage a drawing project: a house, a flower or a rocket.
- Save their best pictures and make a gallery.

🎯 Quick Check

1. Which command lifts the pen?

✅ penup().

2. What angle do you turn for a square?

✅ 90 degrees.

3. What does begin_fill() do?

✅ Starts recording a shape so it can be filled with colour.

4. How many degrees for a triangle?

✅ 120 degrees (360 divided by 3).

5. What does turtle.done() do?

✅ Keeps the drawing window open.